

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2012

EE 206 Digital Electronics & microprocessor

Date: 18.05.2012, Friday

Time: 01:30 p.m. To 04:30 p.m

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1(a) Explain De Morgan's theorem. [04]
- (b) Convert following numbers to it's octal equivalent. [03]
- i) $(1100101011.1110)_2$ ii) 37.29_{10} iii) 672_H
- Q - 2(a) Prove that : [05]
- (i) $(A+A')(AB+ABC') = AB$
- (ii) $ABCD + AB(CD)' + (AB)'CD = AB + CD$
- (b) For the logical expression given below, draw the K-map and obtain the simplified Expression. [05]
- $Y = A'B'C'D + A'BC'D + A'BCD + AB'C'D + ABC'D + ABCD$
- (c) Show that both NAND gate and NOR gate are universal gates. [04]

OR

- Q - 2 (a) Implement the following Boolean expression using an 8:1 multiplexer. [05]
- $F(A,B,C,D) = \sum m(1,3,5,10,11,13,14) + d(0,2).$
- (b) Minimize the following expression and realize using basic logic gates [05]
- $Y = \sum m(0,2,5,6,7,8,10,13,15)$
- (c) Convert the following hexadecimal numbers into their equivalent Decimal and octal numbers. [04]
- (i) $(23867)_{16}$ (ii) $(368170)_{16}$
- Q - 3(a) Write notes on BCD to 7 segment decoders. [05]
- (b) Draw the circuit diagram of a positive level triggered S-R Flip Flop and explain its operation with the help of truth table. [05]
- (c) Determine the output state of a 4bit SIPO shift register after 3 clock pulses if the serial in terminal is held HIGH and the initial contents of the shift register are: [04]
- (i) 1 0 0 1 (ii) 0 0 0 1

OR

- Q - 3(a) Explain Full adder circuit using two half adder circuit. [05]
- (b) Write notes on 2-bit Asynchronous up Counters. [05]
- (c) Give the comparison between CMOS and TTL. [04]

SECTION - II

- Q - 4 (a) Calculate the address line required for interfacing the 16K memory with 8085 microprocessor. [04]
- (b) Assume register B holds 93H and accumulator hold 15 H . Illustrate the results of the Instruction - ORA B, XRA B and CMA. [03]
- Q - 5 (a) Explain the following instructions: [05]
- i) MVI r,data ii) STA addr. iii) INR r
- iv) ANI data v) JC addr.(label)
- (b) Place the four most significant bits of the content of memory location 0040H in to memory location 0041H. Clear the four least significant bits of memory location 0041H. [05]
- (c) Give the comparison between CALL and RET & PUSH and POP. [04]

OR

- Q - 5 (a) What is flag? Draw the format of 8085 flag register and explain in detail. [05]
- (b) Write an assembly language program to find largest no. in a data array. [05]
- (c) Write an assembly language program to find two's complement of an 8-bit number. [04]
- Q - 6 (a) Draw the functional block diagram of 8085 microprocessor. Explain in detail. [07]
- (b) Give the classification of memory and explain in brief. [07]

OR

- Q - 6 (a) Explain Addressing modes of 8085. [07]
- (b) Write a program to generate a continuous square wave with the period of 500 μ s. Assume a system clock period of 100ns. [07]